



Instruction Session 2

SAREO: DATA ANALYSIS & LEARNING METHODS (2026)

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Master: Artificial Intelligence & Engineering Systems

What are we going to do today:

- | | | |
|-----------------------------|---------------|---|
| • Hands-on manual exercises | 15:30 – 15:50 |  |
| • | | |
| • Present first assignment | 15:50 – 16:15 |  |
| • Break | 16:15 – 16:30 |  |
| • Group work | 16:30 – 17:15 |  |

Exercises

1. Consider the following time series: $s = [45, 30, 12, 55, 14, 18, 23, 50, 37, 22, 25, 25, 25, 30, 34, 42, 18, 21, 20, 35]$. What would be the output if you filter this time series with a 3-period moving average filter without padding the data at the boundary?
2. Consider the same time series: $s = [45, 30, 12, 55, 14, 18, 23, 50, 37, 22, 25, 25, 25, 30, 34, 42, 18, 21, 20, 35]$. What would be the output if you filter this time series with a 3 period moving average filter if you use padding?
3. Consider the following sequence data: $s = [-1, -0.2, 0.3, 0.4, 0.25, 0, -0.3, -0.6, 0.2, 0.5]$. This data set is presented to an exponentially weighted moving average filter where the weight of the most recent sample is 0.8. Calculate the filtered signal.
4. Consider the same sequence data: $s = [-1, -0.2, 0.3, 0.4, 0.25, 0, -0.3, -0.6, 0.2, 0.5]$. This data set is presented to another exponentially weighted moving average filter. The filtered signal is the following: $f = [-1, -1, -1, -1, -1, -1, -1, -1, -1, -1]$. What is the value of alpha for this filter?



5ARE0

Academic Year 2026 – 2027

Assignment one

Assessing Bicycle Lane Quality Using Smartphone Sensor Data

Groups of 3 · 20% of Final Grade · Deadline: October 4th, 20:00

Can you bike?

Why Does This Matter?

Cities Are Going Green



Urban cycling is surging. Cities worldwide are investing in bike infrastructure — but lane quality monitoring lags far behind adoption.

The Problem is Real



Potholes, cracks, and rough surfaces cause accidents and deter cyclists. Manual inspection is expensive, slow, and incomplete.

You Already Have the Tools

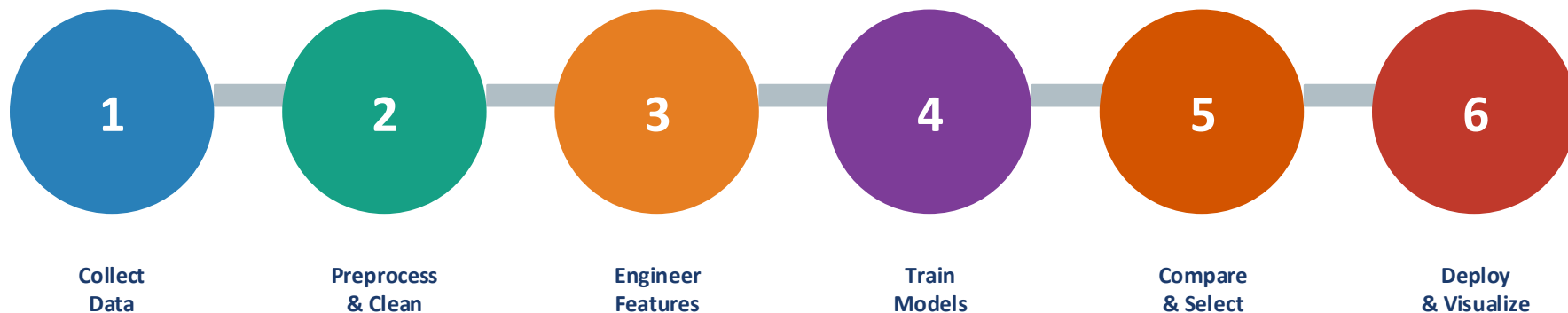


Every cyclist carries a smartphone packed with sensors. This assignment turns that phone into an automated lane quality detector.

This assignment bridges sensing, signal processing, and machine learning — with real urban impact.

Your Mission

Build an automated tool that classifies bike lane surfaces as SMOOTH or BUMPY using smartphone sensor data — from raw signal to deployed model.



💡 The number of recordings, their duration, and the lanes you choose are YOUR decisions — justify and document them all.

4 Supervised Models

4 Unsupervised Models

Your Tools: Smartphone Sensors



Sensor Logger App (iOS/Android, Free) · 100 Hz sampling rate · Standardisation ON · All other sensors OFF

Accelerometer



Captures linear acceleration in 3D (x, y, z). Your primary signal for detecting bumps, potholes, and vibrations as you ride.

Gyroscope



Captures angular velocity and rotational movements. Detects wobbles, swerves, and orientation changes — complementary to acceleration.

Gravity Sensor



Provides device orientation relative to Earth's gravity. Helps normalize readings across different phone placements and riders.

💡 Secure the phone consistently — pocket, armband, or waistband. Document your placement strategy, it affects data interpretation and must be justified.

Data Collection Protocol

Participants

- ✓ Groups of 3 students
- ✓ 2–3 participants collect data
- ✓ All must be course members (5ARE0)
- ✓ All must know how to bike safely
- ✗ Any medical condition preventing biking

Lane Types

SMOOTH

Standard red cycling lanes without imperfections — the baseline for a good surface.

BUMPY

Small potholes, cracks, uneven surfaces — still commonly used, noticeably rough. Not dangerously damaged.

How To Record

Start recording → Pocket the phone → Ride the lane → Stop & save



Safety First — Choose lanes that are uncomfortable but safe. Stop any trial if there is a risk of injury.

Analysis & Modeling Pipeline

3.1 Data Preprocessing

- Handle missing values and sensor noise
- Normalize and standardize signals
- Visualize raw features to guide cleaning

3.2 Feature Engineering

- Handling time series
- Feature extraction
- Feature selection

3.3 Supervised Learning (×4)

- 4 supervised classifiers from course material
- Methods covered up to the deadline
- Choose and justify your evaluation metrics

3.4 Unsupervised Learning (×4)

- 4 unsupervised models from course material
- Compare effectiveness vs supervised approach
- Analyze advantages and limitations

Model Selection & Deployment

Justification Criteria

Problem Suitability

Match model to the classification task

Data Requirements

Appropriate for your dataset size & features

Interpretability

Can end users trust and understand results?

Computational Efficiency

Practical for real-time or mobile deployment?

Performance

Accuracy and metrics across all models

Deployment Task

After finalizing your models:

1. You receive an unlabeled external dataset (NOT for training or testing)
2. Deploy your BEST model on it
3. Visualize which road sections are smooth vs. bumpy
4. This is your real-world product — a live bike-lane quality plot!

What to Submit

 **Deadline: October 4th, 20:00 · Upload to Canvas · 20% of Final Grade**

Submit one ZIP file: **groupX_HAR_submission.zip**

 **groupX.ipynb**

Complete executed notebook · Full pipeline · All code in-house · No .py imports · Re-run before submitting

 **groupX_Report.pdf**

Up to 8 pages · CRISP-DM structure · Tables, plots, and confusion matrices included

 **data/**

Organized raw sensor files · Raw and processed stored separately · Clearly labeled and anonymized

 **README.md**

Setup instructions · Dataset description · Python version and required packages with versions

Include all group member names and student numbers on the notebook title page.



Notebook & Report Details



Notebook Requirements

- End-to-end executable — no .py imports
- All code in-house with clear markdown structure
- Python version + package versions at the top
- Meaningful variable names and comments
- Sections: Load → Preprocess → Features
→ Train → Evaluate → Deploy
- Submitted in EXECUTED form (run it first!)
- Instructors will re-run to validate
- Notebook size: 4 – 50 MB



Report (CRISP-DM, max 8 pages)

Business Understanding

Problem definition & relevance

Data Understanding

EDA, collection methodology, quality

Data Preparation

Preprocessing pipeline & feature strategy

Modeling

Supervised & unsupervised approaches

Evaluation

Metrics, comparison, failure modes

Deployment

Real-world implications & future work

Include tables, plots, confusion matrices, and feature selection visualizations.

Tips for Success



Data Quality First

Good data beats clever algorithms. Collect consistently, label carefully, document your phone placement.



Visualize Before Modeling

Plot every raw signal before touching it. You'll catch noise, glitches, and labeling errors early.



Document as You Go

Note every design decision — why those windows, those features, those models. Your report writes itself.



Iterate Fast

Get a basic end-to-end pipeline running first, then improve each stage. Don't wait for perfect data.



Divide & Conquer

Split roles: data collection, feature engineering, modeling. Sync regularly and review each other's work.



Re-run Before Submitting

Restart kernel → Run All → check every cell. What works in isolation sometimes breaks in sequence.

How we will help you prepare for assignment and exams

- **Canvas > Discussions:** ask and inspire.
- **Instruction sessions.**
 - Implementation of lecture materials.
 - Hands-on exercises to enrich your theoretical understanding.

Group assignment



The elephant in the room



ChatGPT

Writing, brainstorming,
problem solving



Claude

Analysis, long-context
reasoning

Gemini

Research, multimodal
understanding



Codex

Code generation
and execution



Cursor

Coding in editor
(AI autocomplete)



Cursor Agent

Autonomous coding
and task execution



**GitHub
Copilot**

Code assistance
in editor



Perplexity

Web research
with sources



NotebookLM

Work with papers,
PDFs and notes



Grammarly

Grammar, clarity
and writing polish

Overleaf

AI writing assistance
for LaTeX



DeepL

Translation and
language refinement

Canva

Visual design
and graphics



Notion

Organisation, notes
and knowledge management



jupyter

Running and testing
Python code





Go build something that matters.

You're not just completing an assignment.
You're applying machine learning to a real urban problem
that affects every cyclist in your city.

Canvas > Assignments > Assignment 1



October 4th, 20:00 · Canvas Upload

Good luck — we're excited to see what you build! 🎉

Questions? Reach out to your instructors.